

PYTHON FOR PLATFORM ENGINEERS · ISSUE 01 OF 04

Python automation scripts that are safe to run.

If a script can delete, it should first tell you what it
would delete.

01

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Swipe →

Five habits

01 Dry-run by default

Changes only happen with `--apply`.

02 Idempotent

Running twice gives the same result as once.

03 Timeouts and retries

Every network call has both, with backoff.

04 Limited blast radius

`--namespace`, `--limit`, explicit targets.

The skeleton I start with

```
import argparse, logging, sys

p = argparse.ArgumentParser()
p.add_argument("--apply", action="store_true")
p.add_argument("--limit", type=int, default=10)
args = p.parse_args()
logging.basicConfig(level=logging.INFO)

targets = find_stale_resources()[ : args.limit]
for t in targets:
    if args.apply:
        delete(t)
    verb = "deleted" if args.apply else "would delete"
    logging.info("%s %s", verb, t)

sys.exit(0)
```

From laptop to schedule

- 1 Write
- 2 Dry-run in a lab
- 3 Review the output
- 4 `--apply with --limit`
- 5 Full run
- 6 Schedule in a pipeline

The scheduled version runs with a service identity, not your credentials.



Script habits

DO

- ✓ Structured logs
- ✓ Non-zero exit on failure
- ✓ Confirm the target environment
- ✓ Unit-test the selection logic

AVOID

- ✗ Hard-coded credentials
- ✗ Bare except: pass
- ✗ Infinite retries
- ✗ Acting on “all” by default

Before you share it

- ☒ --help explains every flag
- ☒ Dry-run is the default
- ☒ Idempotent on rerun
- ☒ Timeouts on every call
- ☒ Exit codes documented
- ☒ Runs under a least-privilege identity



KEY TAKEAWAY

Default to dry-run, limit the blast radius, log everything.

- --apply to change anything
- Idempotent and time-boxed
- Pipeline identity, not yours

YOUR TURN

What is the most dangerous script in your team's repo?

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